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Integrating Machine Learning for Log Prediction in Seismic Inversion Workflow

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Abstract

The use of machine learning (ML) to predict density and sonic logs from Gamma Ray (GR) logs has demonstrated a stable performance in blocky target reservoirs. Extreme Gradient Boosting is the most appropriate machine learning (ML) algorithm to predict density and sonic logs from Gamma Ray logs. Nevertheless, several iterations are necessary to stabilise the accuracy of the prediction.

Additional data like formation tops and geographical coordinates were added to find a mathematical relationship between GR and density. Unfortunately, it is not possible yet to export the mathematical relationship that links GR to density.

The purpose of the use of ML to predict density and sonic logs was to increase the number of anchor points for the Deterministic Seismic Inversion. That objective was achieved with a high coefficient of correlation and the absolute impedance from the inversion proved to have a better correlation to the previous study from the vintage.

The target is a blocky carbonate reservoir that has an oil-bearing zone at WELL_B and WELL_A with a tilted oil-water contact. That anticlinal trap has 5% porosity cut-offs and no water saturation observed.

The seismic inversion benchmark was useful to prove the concept of additional anchor points using ML log prediction for post-stack inversion. The target reservoir can be more easily mapped using the absolute acoustic impedance.

A large absolute acoustic impedance volume was delivered within a relatively short turnaround time compared to previous legacy data.

Introduction and Objectives

The main purpose of the study is to find the relationship between porosity and acoustic impedance (**Fig. 1**). A post-stack seismic inversion was performed and benchmarked. A Machine Learning method was proposed to predict missing density and sonic well logs to provide more anchor points for inversion.

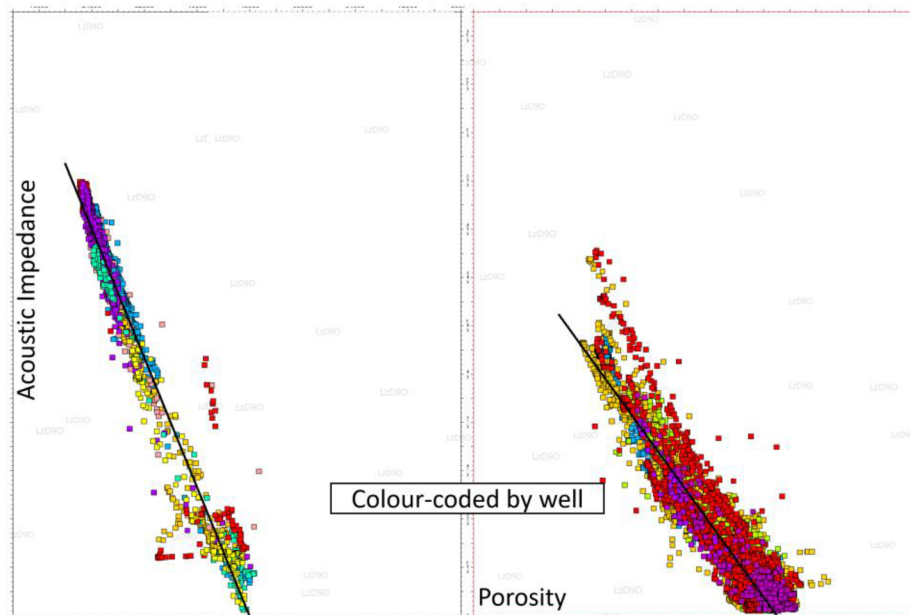


Figure 1—Acoustic impedance vs. porosity (PHIT) at target level (color-coded by well)

Seismic Inversion

Deterministic Inversion

Extracting the Absolute Impedance from a deterministic seismic inversion requires verification of several input parameters. The standard industry practice is to use a PSTM stack with a limited balanced amplitude. Preserved seismic amplitude processing is key to deliver a reliable outcome.

Fig. 1 shows the relationship between porosity and acoustic impedance. A clear linear trend is observed in the target zone, indicating that porosity could be derived from seismic inversion. The trend shall be kept after the use of machine learning to predict acoustic impedance (from predicted density and sonic logs).

The workflow of a Deterministic Seismic inversion is based on wavelet extraction, propagating acoustic impedance from logs, defining the most appropriate low-frequency model, and utilizing the seismic PSTM stack. The seismic data input, combined with the low-frequency model ((Margrave 2011) and (Sharifi, Implications of static low-frequency model on seismic geomechanics inversion 2021)) contribute significantly to the absolute impedance output of the process. Additionally, the wavelet extraction also plays a role, but with a different weighting. The study will not define the weight contribution for each variable in a deterministic seismic inversion. However, the importance of the low-frequency model is crucial ((Alexey Sokolov 2021) and (Wang 2016)), in addition to the correct seismic processing, which would preserve the frequency content and the amplitude variations.

As can be seen, an iterative process can be initiated by minimising the residual between the real and the synthetic data. Minimising the misfits can be achieved by fine-tuning the wavelet extraction and especially by building the most appropriate low-frequency model (Sharifi, Implications of static low-frequency model on seismic geomechanics inversion 2021).

Data Preparation

As part of the data preparation phase, a review of the well logs is conducted for the well that will be contributing to the seismic inversion analysis, with the goal of having logs that are free of spikes and have been fully processed and corrected for any instrumental or geometrical drifts. Cycle skipping in logs due to borehole condition could impact the density or any other acquired logs. Sonic and density logs, in addition to reliable checkshots or VSPs, are the key logs that would be required. Due to business considerations, those

logs are not monitored throughout the entire wellbore. A proposal was made in the study to use Machine Learning (ML) to predict density and sonic logs.

Acoustic impedance was calculated using density and sonic logs in all available wells. A despiked AI was performed (orange dashed line). Residual cycle skipping in the AI log is still present and could impact the propagation, resulting in some impedance anomalies **Fig. 2**.

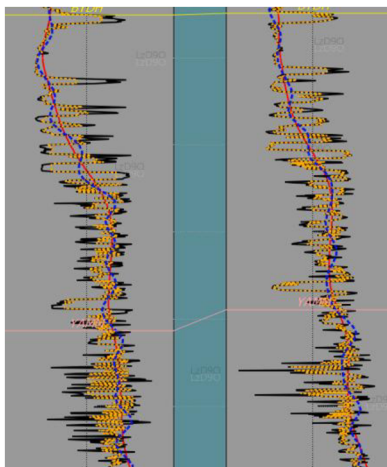


Figure 2—Acoustic impedance calculation from despiked AI

Acoustic Impedance Property Modelling

Acoustic impedance should be propagated in the entire area of interest. Prior to calculating acoustic impedance, a structural time model must be extracted from the seismic data. Relative geological time is derived from the interpretation of the grid model based on the seismic data polarity. The grid-based model was calculated using peak, trough and zero crossing. Positive and negative inflections could be also considered in the iteration run of the grid-based model.

Structurally coherent polarity seismic events from the grid-based model are mapped and saved to build the Relative Geological Time (RGT) model. The RGT model could also be processed using the highest seismic event picked through the grid-based model workflow.

Seismic reflection quality and focusing is highly dependent on the RGT model. It is recommended to tune the input data with post-stack processing. Reducing processing time and better seismic horizons picking are the main targets to build an efficient RGT model.

The target shows the same polarity as the events extracted from the grid-based model, which correspond to a zero-crossing event on the seismic data.

The Relative Geological Time (RGT) model is computed based on the highest seismic polarity. In this study, the seismic polarity of the target seismic reflector is stable and no major phase shift was observed across the area of interest. This indicates that the RGT model is strongly aligned with the conditioned seismic data.

Several algorithms are available to propagate acoustic impedance. Kriging or inverse distance interpolation methods could be utilized. Inverse distance interpolation was used to propagate the acoustic impedance, with the RGT model guiding the distribution of acoustic impedance.

Prior to the property propagation, the Relative Geological Time (RGT) model has to be selected in addition to the chosen seismic data.

Fig. 3 shows the main results of the well property modelling over nearby wells. Some aliasing could be seen in the results, but the propagation of the property remains very reliable. The use of the RGT model, in addition of the seismic data, allows for capturing the variability of the property along its surfaces.

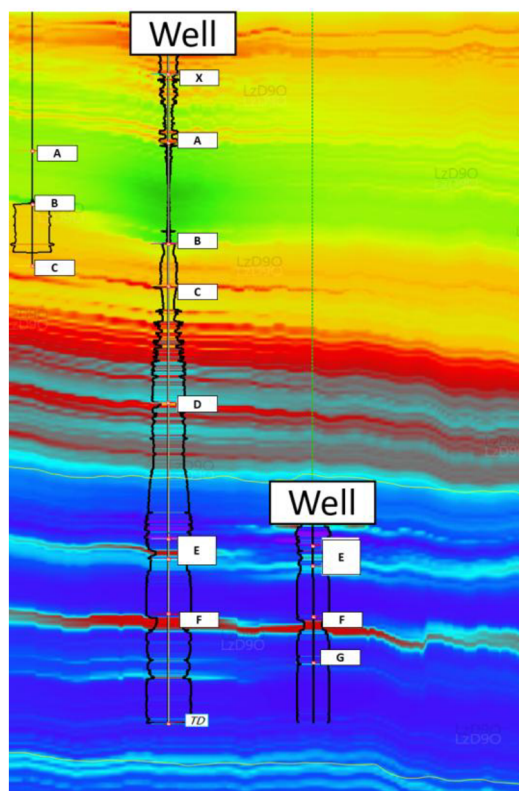


Figure 3—Acoustic impedance propagated across the RGT

Nevertheless, the propagated property could not be used as is, and will instead serve as a guide for the Low Frequency Model.

Low Frequency Modelling

The Low Frequency Model is an a priori model derived from the impedance, from which the Deterministic Inversion process starts. The primary purpose of a Low Frequency Model is to recover the low frequencies from impedance logs and then use this model as a starting point for the deterministic inversion process. The Low Frequency Model in seismic inversion is essential for capturing large-scale geological features.

Unfortunately, low frequencies are often not acquired during seismic acquisition or may be lost during the processing phase. The acquisition parameters indicated that the frequency was reordered using a linear sweep between 5-120 Hz. Multiple references, such as (Margrave 2011), (Sams 2017) (Sharifi, *Implications of static low-frequency model on seismic geomechanics inversion* 2021), and (Nishant Kumar 2012) describe several approaches to building a decent low-frequency model. That key step is necessary to be defined properly to have an anchored start for the seismic inversion. The frequency gaps between seismic acquisition, recording, and processing require the use of well logs to recover those missing low frequencies.

Enhancing the Low Frequency Model is essential, and is accomplished by integrating additional wells. Missing well logs were compensated through Machine Learning predictions of density and sonic values. Acoustic impedance from both original and predicted logs further strengthens the model.

As it can be seen in Fig. 4, the low-frequency model has been populated to the deepest target.

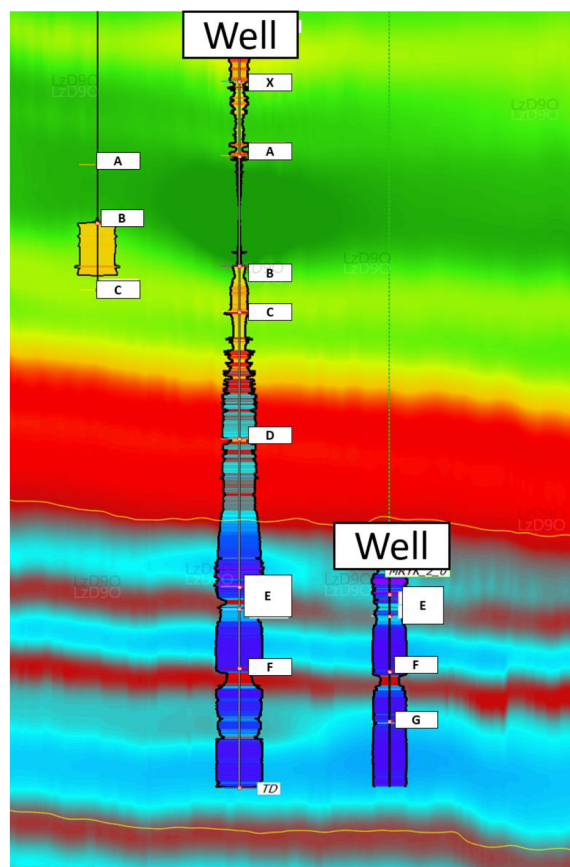


Figure 4—Filtered acoustic impedance at 10Hz (acoustic impedance ranges between 12000 to 65000 g.ft(cm3.s))

Wavelet Definition

Extracted wavelet from the seismic is also another key parameter that drives the deterministic inversion. The wavelet is extracted using the legacy seismic vintage. Wavelet length shall be selected carefully based on the nominal frequency and the selected target. In the study, the main target reservoir was the focus and the wavelet extraction was performed by selecting the most stable wavelet to avoid any minimum phase or inverse polarity issues. **Fig. 5** shows the wavelet shape, frequency, and phase associated with the seismic vintage. The frequency spectrum has values between 20 to 50 Hz, which is also confirmed in other seismic vintages analysed. Frequency lower than 20 Hz must be estimated to have a reliable initial model for the seismic inversion.

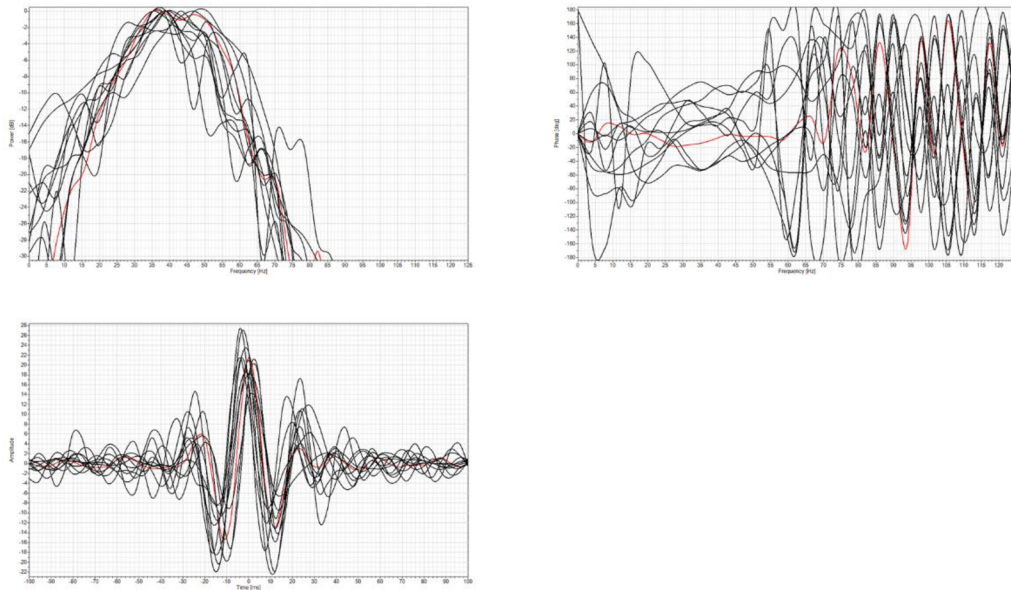


Figure 5—Seismic wavelet extraction for wells (black curve) in addition of the average wavelet (red curve)

Least-Square Inversion

Multiple a priori models could potentially fit the observed data. It is essential to quantify how good the a priori model is and how much freedom will be allowed to minimise the difference between the model and the observed data. It is also critical to understand that a solution exists, but it is not unique, and is prone to instability:

- **Solution Existence:** there may be no model that exactly fits the data. This occurs because of the mathematical approximation of the Least-Squares equation and the presence of noise in the data.
- **Uniqueness:** A non-unique solution is present in an inverse problem.
- **Instability:** The process of computing an inverse solution is unstable, and can give a major change in the estimated model. To mitigate this instability, different regularisation techniques could be considered (Aster 2013).

The Least-Squares method uses a low-frequency filtered a priori model, with a wavelet that is preliminary defined (Menke 1989) and (Aster 2013). The Low Frequency Model, also called a priori model, will be the starting point of the inversion outcome. A percentage of deviation is allowed to provide enough freedom to the a priori model to be influenced by the seismic data. The objective is to minimise the difference between the model and the observed data through an iterative process. Multiple iterations in the parametrisation phase are required to select the model suitable deviation percentage that will be allowed to the priori model.

Outcome Review

Absolute acoustic impedance is the main deliverable from the workflow. Using Machine Learning to predict density and sonic logs from GR has been used to calculate the predicted acoustic impedance (Track #3 in Fig. 9). The absolute acoustic impedance benchmark, which was previously delivered in 2016, has been reprocessed with the latest seismic inversion, as shown in Fig. 7. Only three (3) wells were available: WELL_I, WELL_EOF and WELL_560. Fig. 7 summarizes the benchmark from this current absolute impedance. Fig. 6 shows an example at a well location with a correlation of 94%.

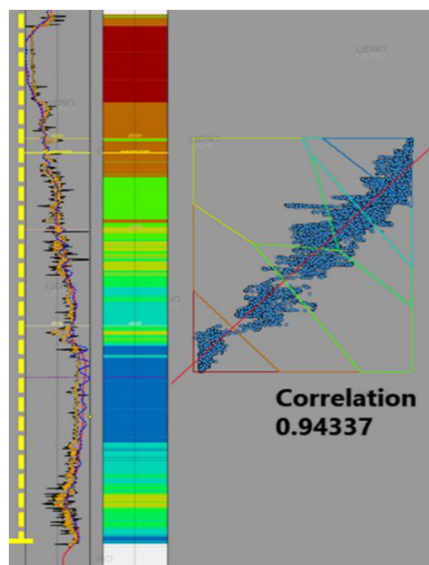


Figure 6—Correlation between acoustic impedance from logs (X axis) and absolute impedance (Y) for WELL_X

A map compilation of the coefficient of correlation is shown in the Fig. 8. Those results summarize the cross plot between acoustic impedance from logs vs. absolute impedance from the inversion. The map shows a high value of correlation.

The previous seismic inversion was heavily impacted by the low-frequency model and is strongly noisy, as illustrated in Fig. 7. The target reflection continuity in the absolute impedance volume is more continuous and sharper, which corresponds to the blocky shape of the reservoir seen in the GR log.

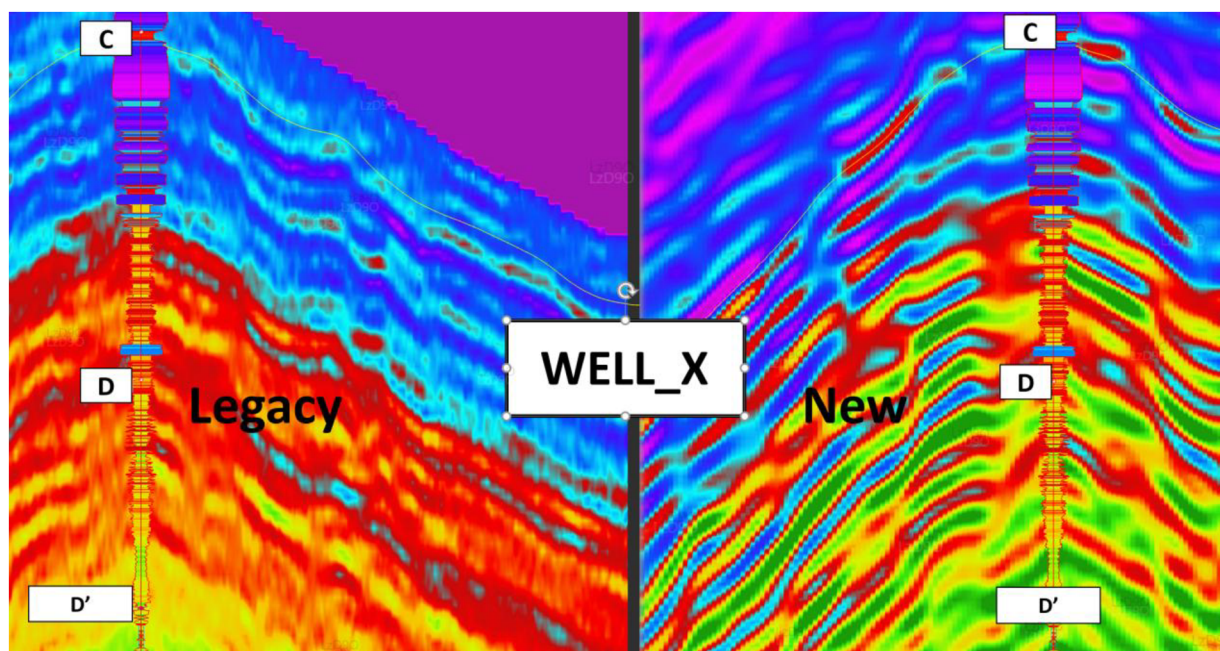


Figure 7—Comparison between legacy and new absolute acoustic impedance at the well location (WELL_X)

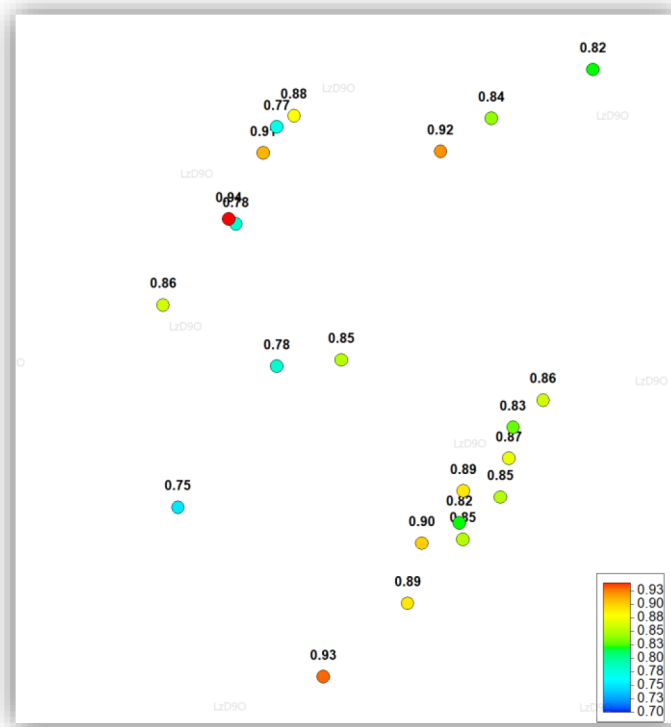


Figure 8—Coefficient of correlation compilation of acoustic impedance vs. absolute inverted data

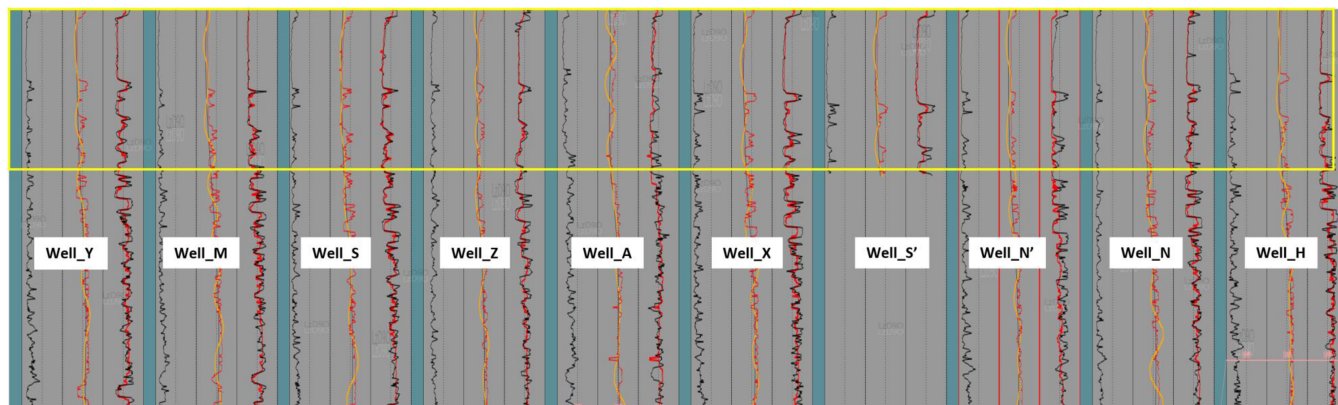


Figure 9—Well tie flattened to target

Track #1 GR;

Track #2 Low-Frequency Model in black, predicted AI from ML in red and Absolute Impedance in orange;

Track #3 Raw acoustic impedance in black and Machine Learning AI from predicted density and sonic logs in red.

Machine Learning

Extreme Gradient Boosting, Neural Network, Random Forest Regressor, Linear Regression, Lasso Regression, Ridge Regression, Least Angle Regression, Lasso Least Angle Regression, Orthogonal Matching Pursuit and Bayesian Ridge are the main algorithms utilized in the study.

The main algorithm that has been used, which provided the most favourable predictions results, was Extreme Gradient Boosting. The main reason for the success of using the XGBoost algorithm in ML is

to the availability of fast GPU computational resources. That reduces significantly the turnaround time, in addition to acceptable validation and generalization accuracies.

It is important to note that there is no dedicated workflow for applying Machine Learning to geoscience.

Extreme Gradient Boosting refers to a class of ensemble machine learning algorithms that can be used for classification or regression predictive modeling problems. Ensembles are constructed from decision tree models. Trees are added one at a time to the ensemble, with each new tree designed to correct the prediction errors made by prior models. This is a type of ensemble machine learning model known as boosting.

Models are fit using any arbitrary differentiable loss function and gradient descent optimization algorithm. This gives the technique its name, "gradient boosting," as the loss gradient is minimized during the model fitting process, much like a neural network.

XGBoost is known for its efficiency, scalability, and high predictive performance. It combines the concepts of gradient boosting and regularization to create a robust and accurate model. Let's break down the main variables and concepts associated with XGBoost:

- **Gradient Boosting:**

Gradient boosting is an ensemble learning technique that combines multiple weak learners (typically decision trees) to create a strong predictive model.

It builds the model in an iterative manner, where each new tree is trained to correct the mistakes made by the previous trees.

The idea is to sequentially fit new trees to the residuals (i.e., differences between the actual and predicted values) of the previous trees, reducing the overall error with each iteration.

- **Decision Trees:**

Decision trees are the base learners used in XGBoost. Each decision tree makes a series of hierarchical decisions based on the input features to arrive at a prediction.

Decision trees are built by recursively partitioning the data into subsets based on the feature values, optimizing some criterion at each split.

In XGBoost, decision trees are typically shallow (with a low max depth) to avoid overfitting and improve efficiency.

- **Objective Function:**

The objective function in XGBoost defines the loss or error measure that needs to be minimized during model training.

The choice of the objective function depends on the specific problem. For example, for regression tasks, the objective function can be mean squared error (MSE), while for classification, it can be log loss (binary or multi-class).

XGBoost provides a wide range of built-in objective functions, and users can also define custom objective functions.

- **Regularization:**

XGBoost employs regularization techniques to prevent overfitting and enhance the model's generalization ability.

Regularization terms are added to the objective function to control the complexity of the model and discourage it from fitting noise or outliers in the data.

The two main regularization techniques used in XGBoost are:

L1 Regularization (Lasso): Adds the absolute values of the weights of the features as a penalty term, encouraging sparsity in feature selection.

L2 Regularization (Ridge): Adds the squared values of the weights of the features as a penalty term, encouraging more balanced weights.

- **Learning Rate:**

The learning rate (also known as the shrinkage parameter or step size) determines the contribution of each tree to the final prediction.

A lower learning rate makes the model more robust to overfitting but requires more boosting iterations to converge.

A higher learning rate accelerates the convergence but makes the model prone to overfitting.

- **Number of Trees:**

The number of trees (boosting iterations) determines how many decision trees are trained sequentially.

A larger number of trees can potentially improve the model's performance but might increase training time and computational complexity.

Determining the optimal number of trees often involves experimentation and using techniques like cross-validation.

- **Hyperparameters:**

XGBoost^[1] has various hyperparameters that control the behavior and performance of the model.

Some important hyperparameters include the maximum tree depth, subsample ratio (sampling rate of data instances), column subsampling ratio (sampling rate of features), and minimum loss reduction required for a split.

Tuning these hyperparameters can significantly impact the model's performance and generalization ability.

XGBoost^[2] has gained popularity in various machine learning competitions and real-world applications due to its ability to handle complex datasets, handle missing values, handle different types of variables, and provide feature importance analysis. It is widely used in both regression and classification tasks and has implementations in popular programming languages like Python and R.

Data Preparation

Density Prediction from GR

GR was the initial log that was used to predict density and sonic logs. The training was considered using also X and Y coordinates, formation tops, and TVD (ft).

A reliable validation and general accuracy were observed to density prediction from GR **Table 1**.

Table 1—Multiple runs using the Extreme Gradient Boosting algorithm for density prediction

ID #	Training Points #	Algorithm	Validation Accuracy	General Accuracy
7329514	442350	XGBoost	0.9266	0.8944
7976547	442350	XGBoost	0.8804	0.8741
6392713	442350	XGBoost	0.9103	0.8779
2689023	442350	XGBoost	0.9156	0.8915
9551548	442350	XGBoost	0.9018	0.8880
5438086	442350	XGBoost	0.9324	0.9074
3732994	442350	XGBoost	0.9298	0.9086

A sensitivity run is available in **Table 2** and demonstrates the higher accuracies when using the XGBoost algorithm.

Table 2—Multiple runs of all available algorithms for RHOB prediction

ID #	Training Points #	Algorithm	Validation Accuracy	General Accuracy
367527	415135	XGBoost	0.9445	0.9042
2157949	415135	Bayesian R.	0.3752	0.3747
357119	415135	Lars	0.3756	0.3677
765399	415135	Lasso	0.3747	0.3829
3226193	415135	Lasso Lars	0.375	0.3786
7748712	415135	Linear Reg.	0.374	0.3864
5805576	415135	MLP Reg.	0.8785	0.8728
1395183	415135	Orthogonal	0.375	0.3792

Extreme Gradient Boosting demonstrated higher accuracy across multiple runs, as shown in **Table 2**.

Sonic Prediction from Density and GR

After fixing the higher runs for density prediction, the sonic log was predicted using the GR and density logs, and other geometrical parameters. **Table 3** shows multiple runs for sonic prediction.

Table 3—Multiple runs using the Extreme Gradient Boosting algorithm for sonic log prediction.

ID #	Training Points #	Algorithm	Validation Accuracy	General Accuracy
6408329	846131	XGBoost	0.826	0.8012
976968	846131	XGBoost	0.7881	0.7929
4738975	846131	XGBoost	0.7496	0.7539
5463584	846131	XGBoost	0.8128	0.8103
2019103	846131	XGBoost	0.7904	0.7878
2186991	846131	XGBoost	0.7362	0.7338

Blind Test

To validate the learning process, it was necessary to run blind testing (**Table 4**). The main assumption used in the blind test was to disseminate the blind tested wells over the target area. It was decided to exclude 10% of the wells for the blind test process. To demonstrate robustness of the learning process, those wells were excluded mainly because they have all type of logs acquired, in addition to being spread over that large area.

Table 4—Blind test prediction runs for RHOB prediction from GR and sonic logs

ID #	Training Points #	Algorithm	Validation Accuracy	General Accuracy
9506129	359791	XGBoost	0.949	0.9350
7058895	359791	XGBoost	0.9459	0.9447
1176506	359791	XGBoost	0.9405	0.9398
601772	359791	XGBoost	0.9471	0.9370
6876855	359791	XGBoost	0.9335	0.9408

Acoustic Impedance Recalculation

As can be seen in the well correlation, the recalculation using the blind test validates a good prediction of the density and sonic logs, which provide a reliable acoustic impedance.

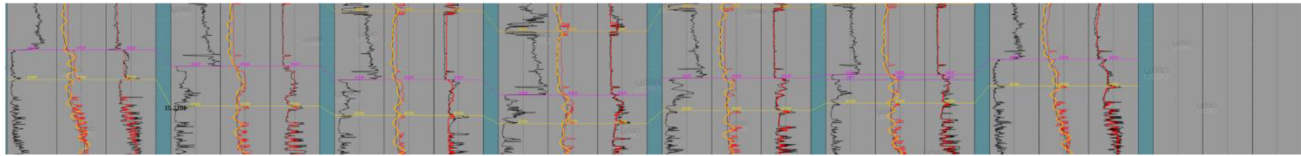


Figure 10—Acoustic impedance recalculation after ML log prediction

Track #1 GR;

Track #2 Low-Frequency Model in black, predicted AI from ML in red and Absolute Impedance in orange;

Track #3 Raw acoustic impedance in black and Machine Learning AI from predicted density and sonic logs in red.

Beyond Seismic Inversion

- Conclusions

Log prediction using Machine Learning could be considered as part of data preparation in case of a limited amount of data (missing sonic or density logs). It has demonstrated that the prediction using the XGBoost algorithm remains very stable using several iterations to stabilise the prediction. A minimum of 5 runs is recommended.

Although not all the parameters in the XGBoost algorithm are fully detailed, it remains the most appropriate algorithm due to its GPU coding.

The seismic data preparation in seismic inversion is mandatory in any kind of seismic environment.

Seismic inversion benchmark was also key to deliver a reliable impedance volume that has good correlation between porosity and impedance at the target level.

A good linear correlation between porosity and acoustic impedance was recognised in the target reservoir.

Machine learning in geoscience is a vast and largely uncharted territory, lacking a well-established workflow. However, the integration of machine learning-predicted logs to enhance low-frequency models has led to notable improvements in seismic inversion outcomes. Despite these advancements, the quality of seismic data remains a critical factor in achieving sharper seismic reflections, particularly at target levels where distinguishing fluids, acoustic impedance, and facies is essential.

- Recommendations

A seismic reprocessing was initiated to emphasize the seismic amplitude anomaly and to better constrain the derived porosity from acoustic impedance.

The time spent in creating the Low-Frequency Model is critical for generating a stable and reliable impedance volume.

Reducing the scattering in the impedance vs porosity plot would lead to a better understanding of the target reservoir.

Machine Learning could be extended to predict losses if the correct input data is properly provided, and it could also be used to provide additional Time-Depth curves for wells that do not have any checkshots.

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